CT PERFORMANCE EVALUATION REPORT

CT - 02 , CELL 02

OWNER / CLIENT: DHARIWAL INFRASTRUCTURE LTD CHANDRAPUR

TESTING AGENCY: SS COOLING TOWER CONSULTANTS

DATE OF TEST: 27 March 2026

REPORT DATE: 15 April 2026

7 Uma Sada – RCHS, CA Road, Chembur, Mumbai – 400071 | www.ssctc.org

Assessment based on CTI ATC-105 Merkel Method

PREAMBLE

Due to poor performance of the cooling tower, factors influencing the poor performance were identified to be progressively resolved. To carry out this procedure it was decided to carry out the changes on one cell and based on the result carry out the same on balance cell. Accordingly one cell was isolated and water from the cell was collected at both ends and the temperature of the collected water was measured. This procedure had to be conducted both before change of gear box and fan and post change of gear box and fan (running at higher power) to ascertain the effect on cold water temperature on one single cell.

Accordingly on 23rd of September 2025, Pre upgrade test was carried out. Testing team reached site and carried out collection of data from 4 to 8 pm. Stabilised peak load was available at 4.0 pm and hence stable readings from 5 to 9 pm were collected. Most stable 1-hour readings are taken up for assessment / benchmarking to be compared with post gearbox and fan replacement measurements.

The existing Maya fan was replaced with another fan and new gear box of higher rating. The fan was operated at 116 kW. Based on these conditions a single cell Post change test was carried out on cell no. 2 in exactly the similar manner to the pre test.

Subsequently distribution system modifications were made and test was conducted on 27 March 2026 in a single isolated cell to understand effect of these changes. The additional head available caused flow to increase substantially in the cell.

MEMBERS PRESENT FOR THE TEST

CLIENT REPRESENTATIVES (DHARIWAL INFRASTRUCTURE LTD CHANDRAPUR):

- Mr Shrikant Shrivastava (Remote)
- Mr GAURAV HANTODKAR

- Mr PAWAN GAWANDE

SS COOLING TOWER CONSULTANTS (SSCTC):

- Mr SURESH SARMA
- Mr SANJAY GORAD
- Mr MRADUL VISHWAKARMA
- Mr PARAG VISHWAKARMA
- Mr RAHUL MANKE

ASSESSMENT METHOD & ANALYSIS

Pre test was conducted on 27 March 2026.

Cell no. 2 has been isolated at the basin level for collection of cold water directly from the rain zone by providing sheets of FRP placed on structural members at the top of the basin so as to collect water from the rain zone and move the water by slope towards the drums placed on either air inlet.

All required data was collected and report has been given.

Since 2 different sets of conditions have to be evaluated, there has to be a reference to which both these test-collected data can be compared. This base has been taken as the design conditions so pre test has been compared to design conditions and post test has been compared to design conditions.

INSTRUMENT PLACEMENT

Air flow was measured using Data Logging anemometer, manually as per CTI ATC-143 Method of equal area — 10 traverses per quadrant, 40 readings total for one fan.

Hot water temperature: taken at inlet of the hot water to the cooling tower, common for all cells.

Cold water: 24 RTD sensors — 12 on Side A and 12 on Side B — measuring temperature of water collected from the isolated cell rain zone.

Water flow: measured using UFM (GE Make) on the riser.

DBT: measured and noted for reference.

WBT (inlet): measured on either side of the air inlet using wet-bulb automatic stations recording WBT every minute.

Fan power: to the fan motor was noted from the client MCC.

CONCLUSIONS AND REMARKS

1. Dhariwal Infrastructure Ltd – Chandrapur had been experiencing shortfall in cooling tower performance.
2. With a view to identify shortfall and prepare an action plan to enable thermal performance improvement, testing based on CTI ATC-105 was carried out.
3. Cooling Tower Performance test conducted on CT2 in March 2025 showed 5.7 °C CWT deviation as per CTI ATC-105 procedure by SSCTC.
4. Probable reasons for the shortfall and modifications for improvement were suggested by SSCTC.
5. With a view to projecting and assessing single-cell performance, benchmarking was carried out (Pre Test, September 2025). Fan replacement showed approx. 1.5 °C improvement (considered 0.7 °C after uncertainties).
6. A second modification — repositioning of the distribution system for better uniform water spray to the fills — was tested on 27 March 2026.
7. ATC-105 analysis of the 27 March 2026 test shows a performance shortfall of 3.80 °C and a capability of 70.4 %.
8. Single cell isolated testing involves UNCERTAINTY in measurement due to turbulent water flow, number of streamlets, and instantaneous temperature changes of streamlets.
9. Considering uncertainties, the expected improvement after changing the distribution system for the full cooling tower is predicted to be approx. 1.1 °C.

SUGGESTIONS & SPECIAL NOTES

1. With 3 pumps operating and one cell under higher flow, velocities through the duct will be higher by ~1.23%. To mitigate, one-size-smaller ferrule (orifice) can be provided at the last 3 pipes of the cell away from the riser.
2. 60 old Paharpur-type nozzles are still fitted and would cause performance loss — particularly at higher heads post-distribution change. Suggest changing all nozzles as soon as possible.
3. V-bar fills offer high resistance to air flow at higher water loadings specifically due to cascade effect. Performance of the tower is likely to further improve once water loading reduces as more cells receive distribution modifications.

FINAL DATA — PRE TEST vs POST TEST COMPARISON

Parameter	Unit	TEST 1 (Pre-Test)	TEST 2 (Fan Change)	TEST 3 (Distr. Change)
Water Flow	M3/hr	2998	3067.21	3680

WBT	Deg.C	25.25	24.22	21.7
DBT	Deg.C	27.52	24.38	33.51
HWT	Deg.C	44.67	43.21	42.13
CWT	Deg.C	35.08	32.89	32.4
Fan Power At Motor	KW	97.04	116.24	117
Fan Air Flow	M3/s	405.97	499	485
Range	Deg.C	9.59	10.32	9.73
Approach	Deg.C	9.83	8.67	10.7
CWT Deviation (Design)	Deg.C	5.8	4.3	3.8
Improvement (vs prev)	Deg.C	—	1.5	0.47

Notes / Mathematical Findings:

1. Improvement of 1.5 °C in TEST 2 is considered as 0.7 °C after accounting for uncertainties.
2. Improvement of 0.47 °C in TEST 3 vs TEST 2 will increase once water loading reduces (currently ~23% higher per cell with distribution change).
3. V-bar fills are specifically sensitive to water loading — higher loading increases pressure drop, shape distortion and cascade effect.
4. Single cell isolated testing involves UNCERTAINTY due to turbulent water flow at point of measurement.
5. SSCTC forecast: after refurbishment of all cells, CWT prediction at 25.5 °C WBT = 32.8 °C. Expected improvement for full CT = 1.1 °C.

AIR FLOW DATA (Anemometer Traverses)

Metric	Value
Average Velocity	4.99 m/s
Total Free Area	92.25 m^2
Total Air Flow	460.0 m^3/s

ATC-105 PERFORMANCE EVALUATION (CTI METHOD)

STEP 1

Table 1 — CWT vs Range at Test WBT = 9.73 °C (Design Range = 10.0 °C)

Computed from performance curves at three water flows and three range conditions.

Range (%)	Range (°C)	CWT @ 90% Flow (°C)	CWT @ 100% Flow (°C)	CWT @ 110% Flow (°C)
80%	8.0 °C	27.407	28.443	29.435
100%	10.0 °C	27.965	29.114	30.209
120%	12.0 °C	28.331	29.558	30.727

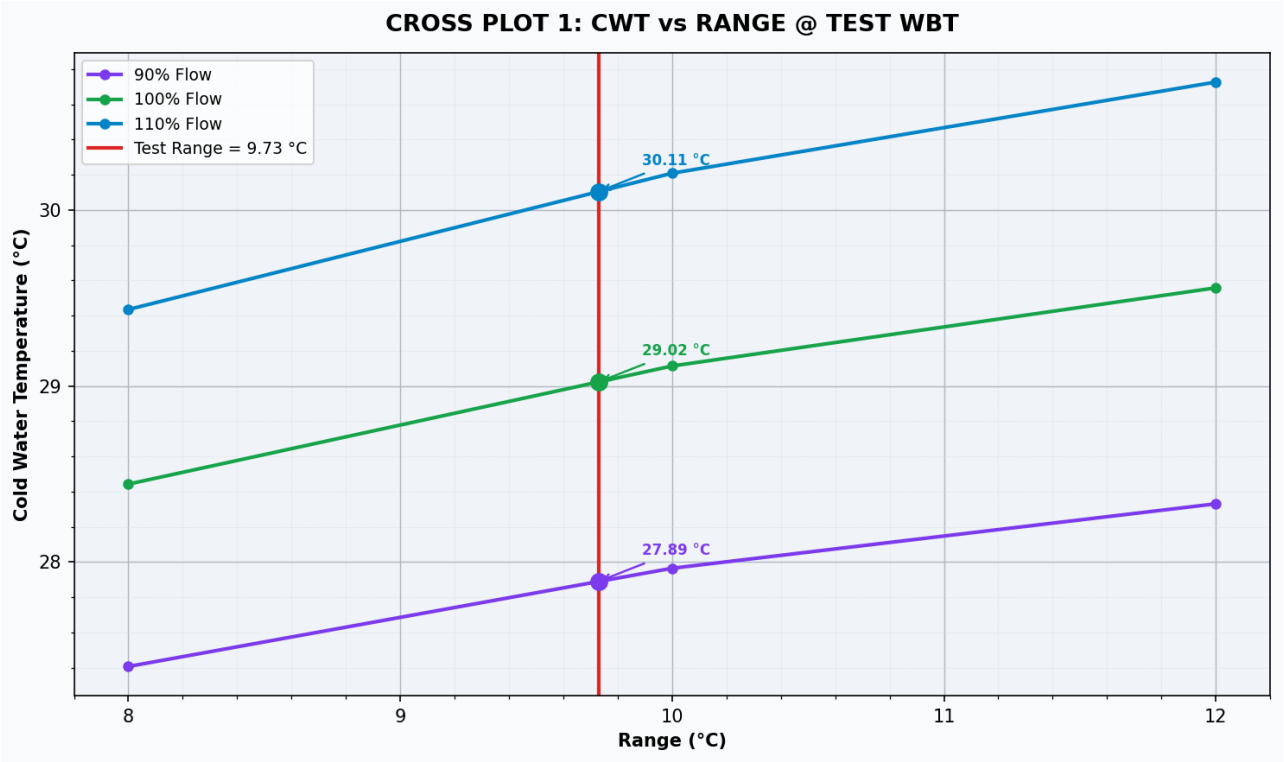
STEP 2

Table 2 — CWT vs Water Flow at Test Range = 9.73 °C

Intersections read from Cross Plot 1 at the test range vertical line.

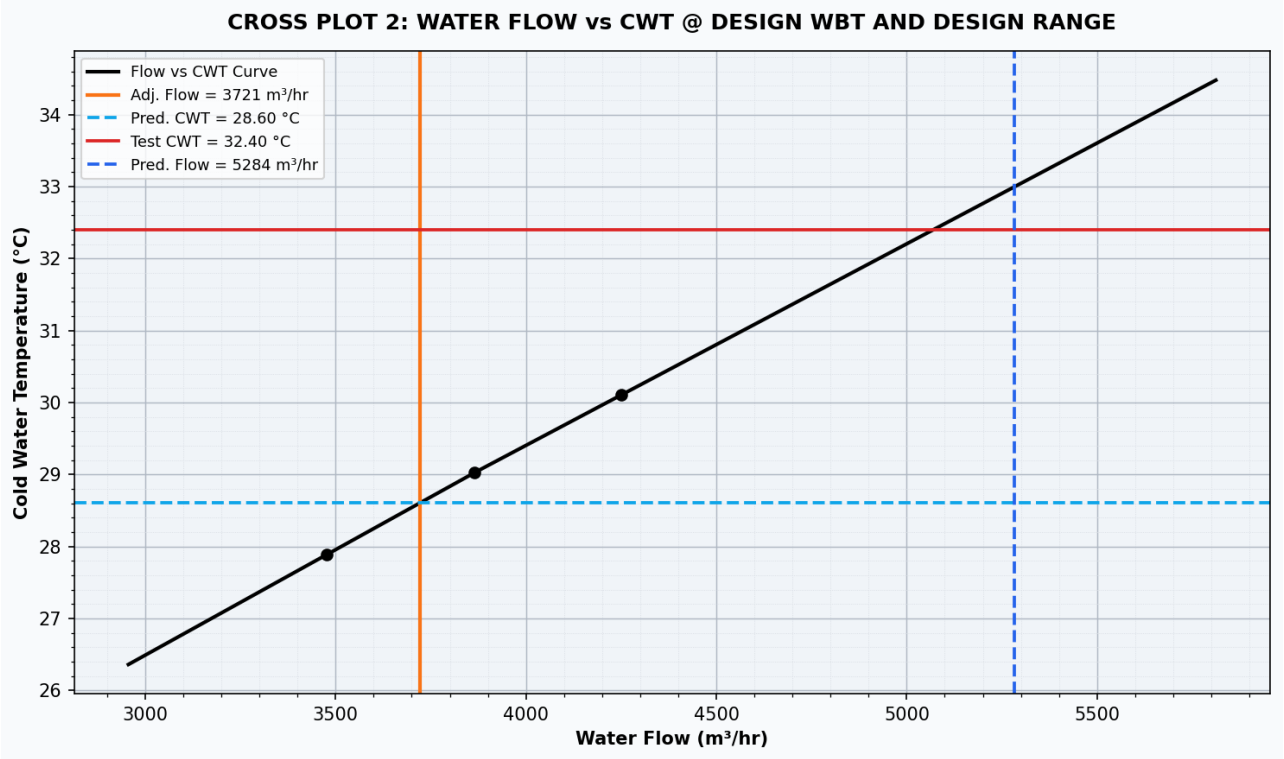
Flow Condition	Water Flow (m^3/hr)	CWT at Test Range (°C)
90% Flow (Purple)	3477.24	27.89
100% Flow (Green)	3863.6	29.023
110% Flow (Blue)	4249.96	30.105

CROSS PLOT 1



STEPS 3 – 5

Cross Plot 2 — Water Flow vs CWT & Final Adjusted Flow Calculation



FINAL ATC-105 MATHEMATICAL RESULTS

Step 4 — Adjusted Water Flow

Item	Symbol	Value	Unit
Test Water Flow (Q_{test})	h	3720.88	Computed ↓
Density Ratio ($\rho_{test}/\rho_{design}$)	i	1.00027	—
Adjusted Water Flow Q_{adj}	—	**3720.88**	m³/hr

Step 5 — Predicted CWT & Performance Shortfall

Item	Value	Unit
Adjusted Water Flow (from cross plot)	3720.88	m³/hr
Predicted (Expected) Cold Water Temp	28.604	°C
Predicted (Design) Water Flow	5283.71	m³/hr
Test Cold Water Temp (Recorded)	32.4	°C

CWT Shortfall: 3.796 °C

Test CWT is higher than Predicted CWT by **3.796 °C**

70.4 %

COOLING TOWER CAPABILITY

(Adjusted Flow / Predicted Flow × 100)